

6(a)(i)	$(I \Rightarrow) 16 / 320 = 0.050 \text{ (A)}$	A1
6(a)(ii)	$R = (25 - 16) / 0.050$ or $R = (9 / 16) \times 320$ or $R = (25 / 0.050) - 320$	C1
	$R = 180 \text{ (}\Omega\text{)}$	C1
	$R_{\text{(LDR)}} = [(1 / 180) - (1 / 240)]^{-1}$ $= 720 \text{ }\Omega$	A1
	or	
	$I = (25 - 16) / 240$ $(= 0.0375 \text{ A})$	(C1)
	$I_{\text{(LDR)}} = 0.050 - 0.0375$ $(= 0.0125 \text{ A})$	(C1)
	$R_{\text{(LDR)}} = 9.0 / 0.0125$ $= 720 \text{ }\Omega$	(A1)

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Question	Answer	Mark
6(a)(iii)	$P = V^2 / R$ or $P = VI$ or $P = I^2 R$	C1
	ratio = $(9^2 / 720) / (9^2 / 240)$ or ratio = $(9 \times 0.0125) / (9 \times 0.0375)$ or ratio = $(0.0125^2 \times 720) / (0.0375^2 \times 240)$ ratio = $0.1125 / 0.3375$ = 0.33	A1
6(b)	resistance of LDR decreases	B1
	resistance of parallel combination decreases or total resistance (of circuit) decreases or current in resistor of resistance 320Ω increases or potential difference across parallel combination / LDR / 240Ω resistor decreases	M1
	voltmeter reading increases	A1